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Large language model agents in a collective-risk social dilemma: cooperation tracks prompt salience, not catastrophe risk

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Collective-risk social dilemmas, of which climate change is the archetype, reward a group only if its pooled contribution crosses a threshold before a probabilistic catastrophe. In the canonical human experiment, cooperation rises steeply with the stated risk. As large language models (LLMs) are increasingly deployed as autonomous agents and as surrogates for human decision-makers, we ask whether they reproduce this risk sensitivity, and, crucially, whether any deviation reflects a genuine behavioural disposition or a mere failure to understand the game. We ran seven instruction-tuned models spanning three families and 7–72 billion parameters as six-agent groups in a replication of the collective-risk dilemma, crossing catastrophe risk ($p = 0.1, 0.5, 0.9$) with language (English, Vietnamese). No model reproduced the human pattern: target-reach was identical at every risk level for all seven, whereas human groups reach the target in 0%, 10% and 50% of cases as risk rises. Because reach is at ceiling for six of seven models, we analyse contributions, and because the design uses common random numbers we analyse them paired: risk then does move behaviour, significantly in three of seven models, but by at most 8.8 of a possible 240 across the entire risk range. Set against this, a manipulation that carries no incentive information at all - printing the running pool total in the prompt - moves the same quantity by up to 95.3 points. Prompt salience thus dominates catastrophe risk by an order of magnitude. In-situ comprehension probes locate the boundary of what the agents can do: they recite the rules (93.8–100%) and read printed values near-perfectly, but summing the pool across rounds fails in small models and recovers with scale (14% to 88%). We stress a limitation that bounds all such studies, ours included: our prompt names the equal-split solution, and several models lock onto exactly it, so what looks like a cooperative disposition is partly compliance with a supplied focal point. We argue that behavioural claims about LLM agents must be tested against prompt-salience controls before they are attributed to preferences.

The prevention of dangerous climate change is a collective-risk social dilemma: many actors must jointly invest enough to keep a shared catastrophe from occurring, yet each actor is individually tempted to free-ride on the investment of others [1–3]. In laboratory social dilemmas, humans sustain cooperation through reciprocity and costly punishment and respond to the material stakes at hand [4]. Laboratory versions of this dilemma have a sharp empirical signature. When Milinski and colleagues had groups of six people play a ten-round threshold game in which failing to reach the target triggered the probabilistic loss of everything they had kept, the fraction of groups that reached the target rose steeply with the announced risk of loss - from essentially zero when the risk was low to about half when the risk was high [1]. Humans, in other words, cooperate more when more is at stake.

1. Introduction

Large language models (LLMs) are now used both as autonomous agents that negotiate, plan and transact on a user's behalf, and as inexpensive surrogates for human participants in the social and economic sciences [5–7]. A growing literature places these models inside classic games and finds that they can cooperate, reciprocate and punish, but often in ways that diverge from human play, and not always in the same direction: some models are strikingly prosocial and forgiving in repeated dilemmas [8, 9], whereas others retaliate unforgivingly after a single defection [10] or fail to sustain a shared resource without institutional scaffolding [11]. If such agents are to be trusted in, or used to forecast, high-stakes collective decisions, we need to know how they behave when the dilemma is structured by *risk* rather than by a fixed payoff matrix, because it is precisely the response to risk that gives the human collective-risk dilemma its policy relevance.

This question is entangled with a methodological one that recurs throughout the study of LLM behaviour. When a model deviates from the human pattern, the deviation admits two very different readings. It may be a genuine behavioural disposition – the model “understands” the situation and nonetheless chooses differently – or it may be an artefact of comprehension, in which the model has simply misread the rules, the history or the current state. These two readings have opposite implications for whether scaling, prompting or fine-tuning will change the behaviour, yet most behavioural studies of LLMs do not measure comprehension directly and therefore cannot distinguish them. Recent work on the prisoner's dilemma has begun to close this gap by interleaving behavioural play with a battery of understanding checks, asking the model to report the rules, the game history and the current state, and scoring the answers against ground truth [8].

Here we bring both questions together for the collective-risk social dilemma. We instantiated the Milinski experiment with LLM agents, replacing each of the six human players with an instruction-tuned model, and swept the announced catastrophe risk across the same three levels used with humans. To ask whether behaviour is bounded by capability, we placed seven models spanning three families and an order of magnitude in size, from 7 to 72 billion parameters, in the identical protocol. To separate disposition from comprehension, we adapted the understanding-check methodology of Fontana et al. [8] from the two-player prisoner's dilemma to the N -player threshold game, probing each model *in situ* on the very prompts from which it was making decisions, with all answers graded against the game engine's ground truth. Finally, because these models are deployed multilingually and the collective-risk framing has an intrinsic numerical structure, we ran every condition in both English and Vietnamese.

Three findings organise the paper. First, no model reproduces the human risk response: target-reach is flat in p for all seven, and although a correctly paired analysis shows that risk does shift contributions in three models, the largest such shift is 8.8 of 240 – far too small to change an outcome. Second, a manipulation with no incentive content whatsoever, printing the running pool total, moves contributions by up to 95.3 points in the same models: the agents are far more responsive to what the prompt makes salient than to what the game makes costly. Third, comprehension probes bound what this can be attributed to. The agents recite the rules and read printed quantities near-perfectly; the one genuine cognitive gap, summing the pool across rounds, is a small-model deficit that closes with scale. Throughout we are careful about a threat that applies to this literature generally and to our design in particular: our prompt names the equal-split solution, and some models reproduce it exactly, so cooperative-looking behaviour cannot be cleanly separated from compliance with a supplied focal point.

2. Material and methods

(a) The collective-risk social dilemma

We reproduced the parameters of the canonical human experiment [1]. Six players each began with an endowment of €40. Over ten rounds, each player privately and simultaneously chose to contribute 0, 2 or 4 to a common pool. If the pooled total reached the target of €120 by the end of the game, players kept whatever they had not contributed; if the target was missed, then with probability p – the catastrophe risk – every player lost all remaining private money. The maximum achievable pool is $6 \times 4 \times 10 = 240$, so the target of 120 can be met if players contribute, on average, half of the maximum. We used the three risk levels of the original study, $p \in \{0.1, 0.5, 0.9\}$. For humans, the fraction of groups reaching the target rose with risk, from 0% at $p = 0.1$ to 10% at $p = 0.5$ to 50% at $p = 0.9$ [1]; this monotone increase is the behavioural benchmark against which we compare the models.

(b) Language-model agents and the game loop

Each human player was replaced by an instruction-tuned LLM. We built a configuration-driven engine, in the style of agent-based game frameworks for LLMs [12], in which the six agents in a group were homogeneous (the same model) and anonymous (neutral identifiers, no assigned personas). On each round, every agent received a natural-language prompt containing the rules of the game and the complete round-by-round history, and was asked to end its reply with a line

of the form `CONTRIBUTION: c` with $c \in \{0, 2, 4\}$, which we parsed deterministically. Agents therefore had full-history memory and a neutral framing that neither encouraged nor discouraged cooperation. One property of this prompt matters for every interpretation that follows and we state it plainly: in describing the target it also names the equal-split solution, telling the agent that €120 corresponds to “an average of 2 per player per round” and adding the worked example that contributing 2 every round leaves 20 at the end. The prompt therefore supplies a risk-independent focal action. It is faithful to the information a human subject could trivially compute, but it means that an agent contributing 2 may be solving the dilemma or may simply be following the number in front of it, and our design cannot separate the two. We return to this in §4. Catastrophes were resolved by a single end-of-game lottery. That lottery is the weakest part of the implementation. It draws one uniform variate per repetition, seeded from the repetition index alone, so the same ten draws are shared across every model, language and risk level: the study contains ten independent lottery outcomes in total, not one per game. All ten happen to fall below 0.5 (mean 0.276), so at $p = 0.5$ and $p = 0.9$ every game that missed the target was recorded as a catastrophe. Realised catastrophe rates are therefore an artefact of the draw schedule and we exclude them from all inference; they are reported only descriptively. Models were served locally with vLLM at a 4096-token context and decoded at temperature 0.7 with a 512-token cap. Sampling was seeded per turn from the experiment seed, the repetition, the round and the agent index, so that runs are reproducible and independent across repetitions while using common random numbers across the risk and language conditions, isolating the manipulated factor. The structure of the decision prompt is summarised in figure 1(a).

We studied seven models spanning three families and roughly an order of magnitude in size: Qwen2.5 at 7B, 32B and 72B [13]; Gemma-2 at 9B and 27B [14]; and Llama-3.1 at 8B and 70B [15]. The two 70–72B models were served with activation-aware weight quantisation. Each model played the full factorial of three risk levels by two languages with ten independent repetitions, giving 60 games per model. Contributions were parsed without failure for all seven models (parse-failure rate 0%), so every game contributed valid data.

(c) In-situ comprehension probes

To test whether behaviour was bounded by understanding, we adapted the meta-prompting battery of Fontana et al. [8] from the two-player prisoner’s dilemma to the N -player collective-risk game. Twenty questions were organised along three axes: *Rules* (the fixed structure of the game - available actions, endowment, target, number of rounds and, critically, the catastrophe probability), *Time* (reading specific past events from the history) and *State* (the current cumulative situation, including the running pool total). Every question had a closed-form ground truth computed by the game engine, so grading involved no second model. Probes were issued on the identical prompts from which the agent was deciding and were appended after the decision, so a probe cannot influence the decision it accompanies. The comprehension runs were nevertheless a separate execution of the game, using greedy decoding for a deterministic benchmark, and their behavioural trajectories therefore differ from those of the main behavioural runs; the two studies characterise the same agents and the same prompts, not the same game trajectories. Because the shared pool total is the quantity most likely to expose a difference between reading and computing, we ran a controlled A/B manipulation in which the running total was either hidden (the default) or printed in the prompt. This manipulation adds no information about incentives - the total is a deterministic function of a history the agent already has - so it serves as a salience control for the behavioural study as well as a comprehension probe, and we analyse its behavioural arm in §(d). It was run for the five models in the comprehension panel at 7–32B. Probes used greedy decoding for a deterministic benchmark and yielded approximately 54,500 graded probes per model. The comprehension panel used the same models as the behavioural study except that the 70B Llama was the Llama-3.3 revision rather than Llama-3.1; we keep the labels distinct throughout. Unparseable answers were graded as incorrect; the probe parse-failure rate was below 0.01% for six of the seven models and 1.03% for Llama-3.1-8B, where it was almost entirely confined to Vietnamese (2.05% of that model’s Vietnamese probes). Accuracies are reported with Wilson 95% confidence intervals [16]. These treat probes as independent, which they are not: probes are clustered within games, rounds and questions, so the intervals we plot are anti-conservative and should be read as a lower bound on uncertainty. Because the comprehension differences we rely on span tens of percentage points, this does not affect any conclusion drawn here, but it would matter for finer comparisons.

(d) Analysis and dependent variables

Our primary behavioural dependent variables were the target-reach rate (the fraction of games in which the pool met €120) and the group contribution (the pooled total, out of 240). We treated the realised catastrophe rate as secondary, because a single end-of-game lottery per game, with the draw shared across languages by the common-random-number scheme, yields only about ten effective draws per model and risk level and is therefore noisy. For the comprehension study the dependent variable was answer accuracy against the engine ground truth. All quantities were recomputed directly from the raw per-game and per-probe logs.

(a) A collective-risk dilemma, played by six LLM agents



(b) Two studies on the same games

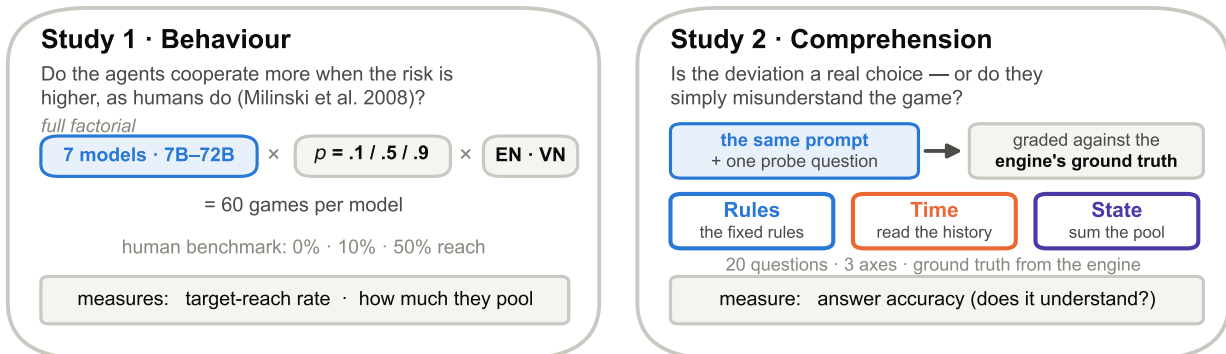


Figure 1. Study design. (a) Six LLM agents play a faithful replication of the Milinski collective-risk dilemma: each of six agents starts with €40 and, over ten rounds, privately contributes 0, 2 or 4 to a common pool from a prompt containing the rules and the full history; at the end, if the pool has not reached €120 the group suffers a catastrophe with probability $p \in \{0.1, 0.5, 0.9\}$. (b) The same games feed two studies. Study 1 (behaviour) crosses seven models (7–72B) with three risk levels and two languages (60 games/model) and measures the target-reach rate and how much the agents pool. Study 2 (comprehension) appends a graded question to the identical prompt, scoring twenty questions along three axes (Rules, Time, State) against the engine's ground truth, to ask whether any deviation is a genuine choice or a misunderstanding.

3. Results

(a) No model reproduces the human risk response

Humans in the collective-risk dilemma cooperate more when the catastrophe is more likely; the models did not. Every model reached the target at the same rate at every risk level (figure 2(a)), against a human benchmark that climbs from 0% to 50% across the same range. Six of the seven models reached the target in essentially every game regardless of risk (Qwen2.5-7B, Llama-3.1-8B, Gemma-2-9B, Gemma-2-27B and Qwen2.5-72B at 100%; Qwen2.5-32B at 96.7%), and the seventh, Llama-3.1-70B, reached it in exactly half of its games at each of $p = 0.1, 0.5$ and 0.9 .

This flatness is easy to describe and hard to test, and we are explicit about the limitation. Five of the seven models reached the target in all sixty of their games, so their reach rate has no variance and admits no regression on risk at all; a sixth varies only between 95% and 100%. Reach is thus censored at the ceiling for six of seven models, and a positive risk response of the human size would be arithmetically undetectable in it. We therefore treat reach as descriptive and take group contribution, which is free to move in either direction, as the informative dependent variable.

Contributions must be analysed paired, not pooled. The design reuses the same sampling seeds across risk levels, so a game at $p = 0.1$ and its counterpart at $p = 0.9$ share their random draws; the resulting within-pair correlations range from 0.32 to 0.95. Treating the three risk levels as independent samples, as a naive analysis would, discards that pairing and badly understates the precision available.

Analysed correctly, risk does move contributions. Every one of the seven models has a positive risk slope, and a paired t -test on the difference between $p = 0.9$ and $p = 0.1$ is significant for three of them: Gemma-2-9B (+8.8 points, $P = 0.0023$), Qwen2.5-72B (+3.2, $P = 0.0062$) and Gemma-2-27B (+1.9, $P = 0.048$); the first two survive Bonferroni correction across

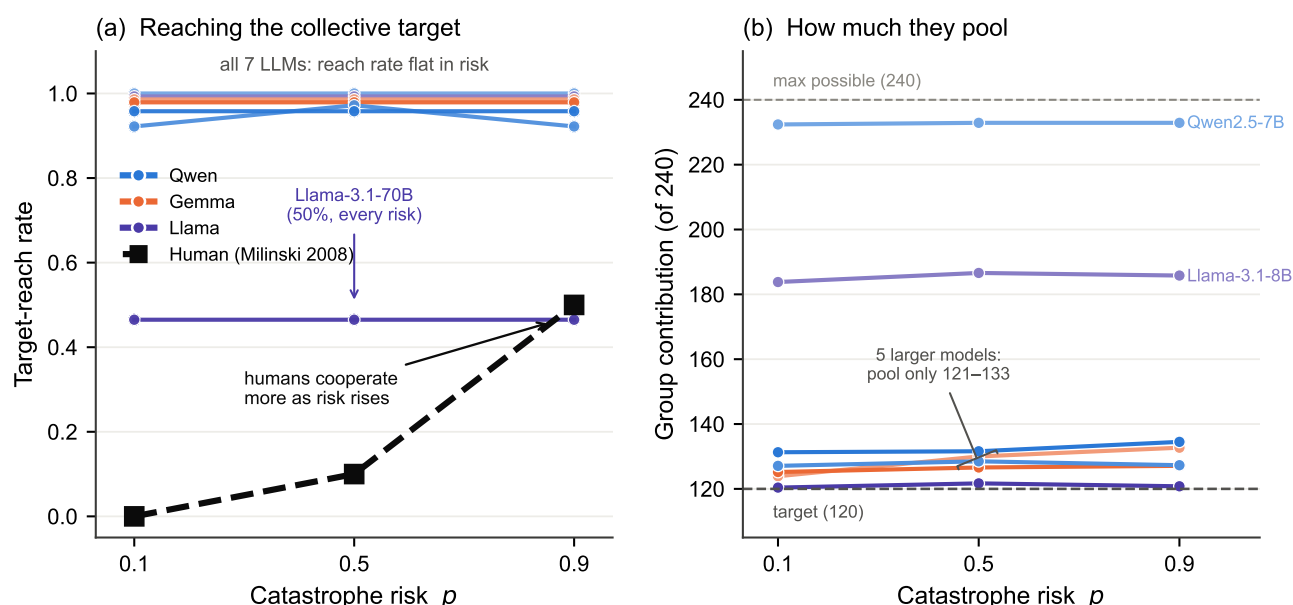


Figure 2. No model reproduces the human risk response. (a) Target-reach rate against catastrophe risk for all seven models (colour denotes family; shade denotes size) and for humans (dashed). Every model's reach rate is flat in p ; humans rise from 0% to 50%. Llama-3.1-70B is flat at 50%. (b) Group contribution (of a possible 240) against risk. Contributions are essentially flat in p (the one weak exception, Gemma-2-9B, is quantified in the text); the vertical spread reflects how economically each model plays, not risk sensitivity. In (a) five models sit exactly on 100%, so the curves are drawn with a small downward offset (at most 0.04) purely so that all seven remain visible; no curve is drawn above its true value.

the seven models. The response is real. What it is not is consequential. The largest effect any model shows across the entire risk range is 8.8 of a possible 240 - about one extra contribution of 2 from one agent in four of the sixty games - and it changes no outcome, since Gemma-2-9B already reached the target in every game at every risk level. Against the human benchmark, where the identical manipulation moves the fraction of successful groups from none to half, this is a response in the right direction and of the wrong order of magnitude.

One caveat about that effect deserves the same treatment we give Llama-3.1-70B below. Gemma-2-9B's response is entirely Vietnamese: in English it contributed exactly 120.0 in all thirty games, a hard constant with no variance and no slope to estimate, while in Vietnamese it rose from 127.8 to 145.4 (slope +22.0, $P = 0.0042$). Two of the fourteen model-by-language cells are constants of this kind, and Qwen2.5-7B sits at the 240 ceiling in 16 of its 60 games. Both dependent variables are censored somewhere in this panel, and no analysis can recover a risk response from a cell with no variance.

(b) Scale buys economy, not risk sensitivity

Although risk did not move the models, scale did move how much they cooperated. In every family the largest model pooled far less than the smallest, though the decline is not strictly monotonic (figure 3). The smallest models over-contributed massively: Qwen2.5-7B pooled 232.7 of a possible 240, close to the ceiling, and Llama-3.1-8B pooled 185.4. Every larger model converged toward the minimum that clears the threshold, pooling between 121 and 133: Qwen2.5 fell from 232.7 at 7B to 127.6 at 32B before ticking back up to 132.5 at 72B, and Llama fell from 185.4 at 8B to 121.0 at 70B, while Gemma was already economical at 9B (128.9) and stayed there at 27B (126.3). In effect, the panel separates into three tiers - a ceiling-hugging Qwen2.5-7B, an intermediate Llama-3.1-8B, and a cluster of five larger models that contribute just enough to clear €120 and keep the rest. Crucially, this economy is orthogonal to risk: the more economical models are not more risk-sensitive, they are simply better at extracting private surplus while still clearing the collective target.

(c) The one model that fails, fails by language

Llama-3.1-70B is the only model that did not reach the target reliably, and it is worth being precise about why, because its pooled average is misleading. A mean contribution of 121.0 appears to place it exactly on the 120 target, as though it were playing a knife-edge strategy. It is not: that average is a mixture of two well-separated modes. In English it pooled 130.3 on average, with every game between 124 and 140, and reached the target in all thirty games; in Vietnamese it pooled 111.6, with every game between 108 and 118, and reached the target in none. Only five of its sixty games fall anywhere in the interval 115–125. Its 50% reach rate is therefore a language split, not a marginal case, and we return to it in §(g).

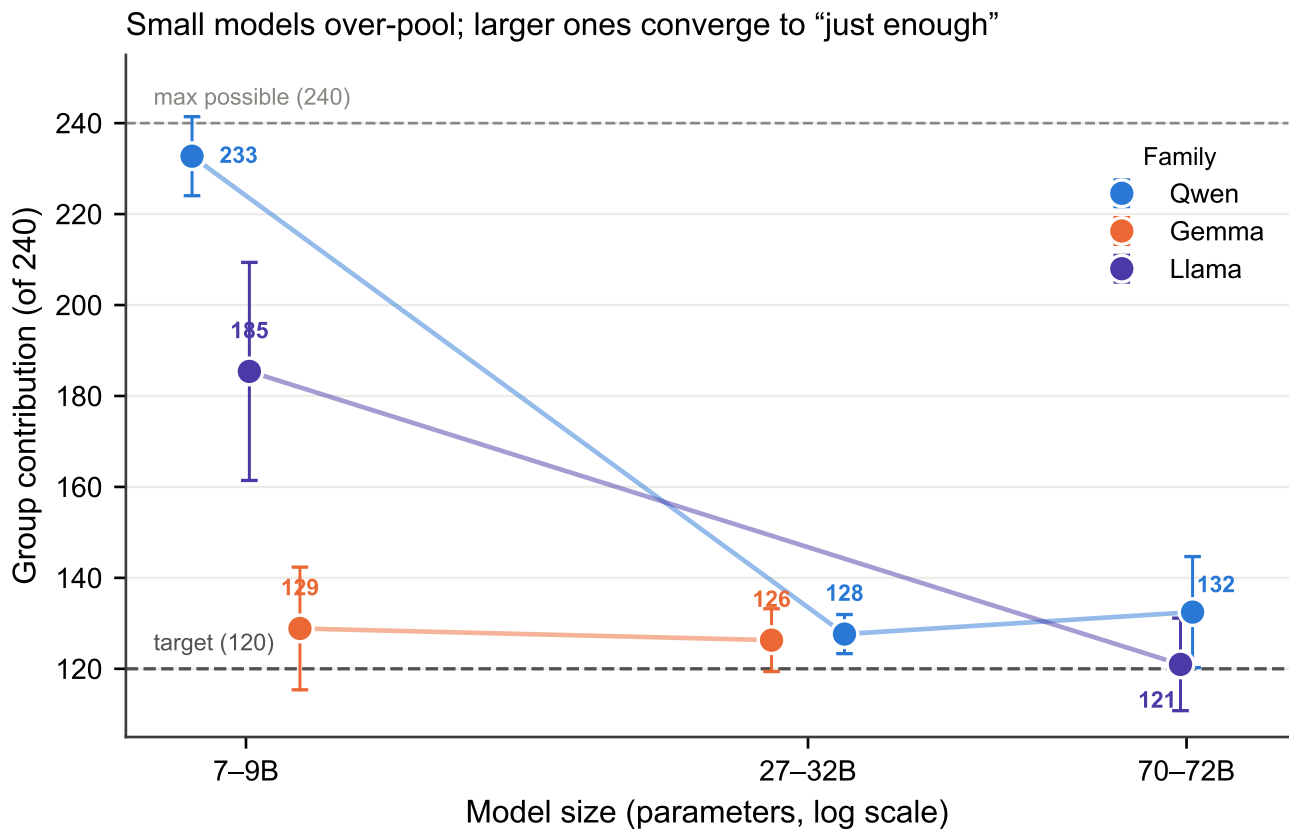


Figure 3. Small models over-pool; larger ones converge to “just enough”. Group contribution (of 240) against model size on a logarithmic axis, connected within family. In every family the largest model pools far less than the smallest, but the decline is not strictly monotonic: Qwen2.5 falls to 127.6 at 32B and rises again to 132.5 at 72B. Error bars are standard deviations across games.

Because its Vietnamese games always missed the target, this model absorbed 22 of the 23 catastrophes suffered in the whole study; the only other model ever caught was Qwen2.5-32B, once. Conditional on having missed the target, its catastrophe rate was 0.2, 1.0 and 1.0 at $p = 0.1$, 0.5 and 0.9, against nominal rates of 0.1, 0.5 and 0.9. With only ten independent lottery draws per cell (§2.4) these realised rates are far too noisy to read as a risk response, which is precisely why we do not treat catastrophe as a dependent variable. The behaviourally meaningful point is that punishment changed nothing: within each language its contribution was flat across risk, so the one model that was repeatedly ruined did not respond by cooperating more.

(d) Prompt salience moves the agents far more than risk does

If tripling the catastrophe risk barely moves these agents, what does? The A/B arm of the comprehension study answers this directly, and it was run on the same games with the same decoding, so the comparison is internal. Printing the running pool total in the prompt adds no information about incentives: the total is a deterministic function of a history the agent is already shown, and it changes nothing about payoffs, target or risk. It nevertheless moves behaviour by an order of magnitude more than risk does (figure 4). Contributions fell by 95.3 points for Qwen2.5-7B ($232.5 \rightarrow 137.2$, paired $P = 6 \times 10^{-34}$), by 26.6 for Llama-3.1-8B ($P = 2 \times 10^{-17}$), by 8.2 for Gemma-2-9B ($P = 2 \times 10^{-6}$) and by 2.9 for both Gemma-2-27B and Qwen2.5-32B. Gemma-2-27B's target-reach rate fell from 100% to 67%. The two smallest models, which over-contributed most when the total was hidden, are precisely the ones that correct hardest when it is shown.

The contrast is the central result of this paper. The manipulation that the game says should matter - a ninefold change in the probability of losing everything - moves contributions by at most 8.8 points. A manipulation that carries no incentive information at all - writing a number the agent could in principle have computed - moves them by up to 95.3. Whatever governs these agents in this dilemma, it is closer to what the prompt puts in front of them than to what the game makes costly.

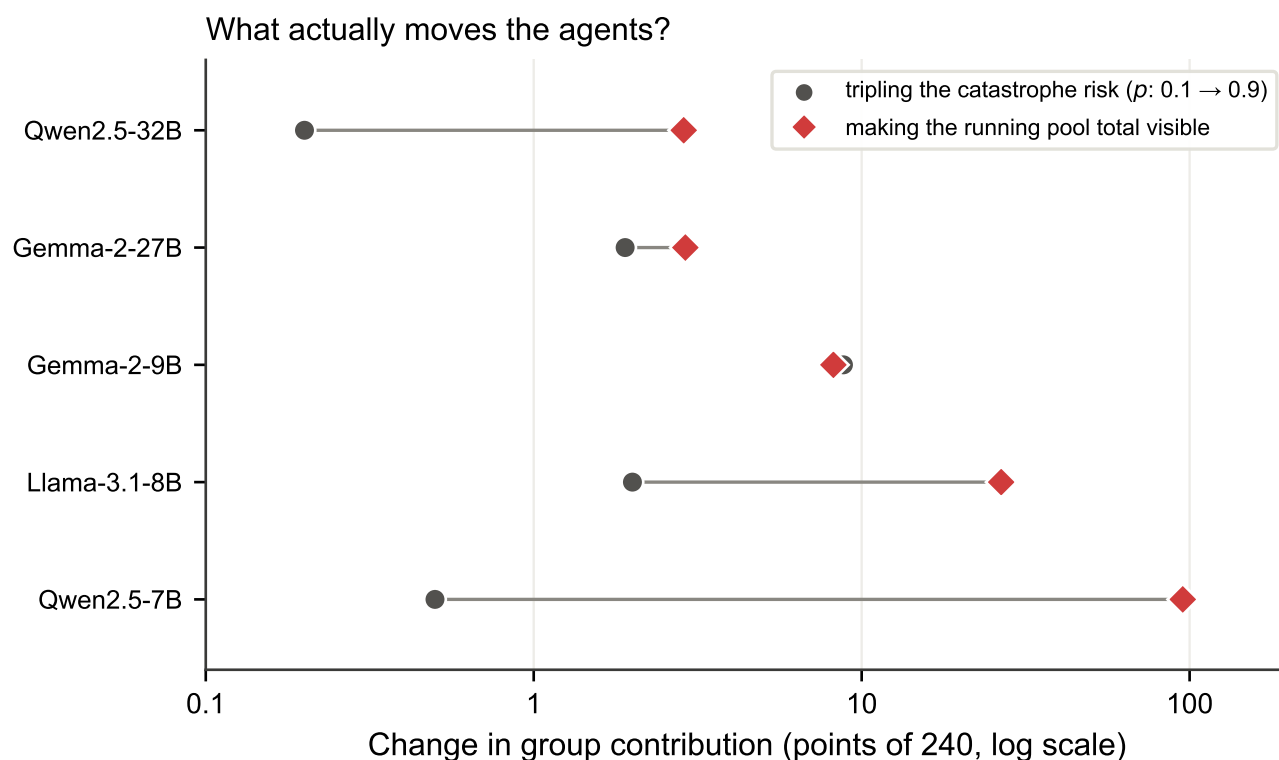


Figure 4. Saliency beats incentive. Absolute change in group contribution (of 240, log axis) produced by two manipulations, for the five models run in both arms: tripling the catastrophe risk from $p = 0.1$ to $p = 0.9$ (grey circles, paired within seed) versus printing the running pool total in the prompt (red diamonds). The saliency manipulation carries no information about payoffs, target or risk, yet dominates for four of the five models.

(e) What the agents can and cannot do

The behavioural deviation might, in principle, be a failure of comprehension: perhaps the models over-cooperated and ignored the risk because they did not grasp the rules or could not read the game. The in-situ probes rule this out. Accuracy on the *Rules* axis was uniformly high, from 93.8% for Llama-3.1-8B to 100% for the five largest models, and accuracy on the *Time* axis - reading specific events from the history - ranged from 82% to 99% (figure 5). One probe asked the model to state the catastrophe probability of the game it was playing, and every model answered almost always correctly: 98.1% for Qwen2.5-7B, 91.4% for Llama-3.1-8B and 100% for the other five (figure 6(b), upper curve). We are deliberately modest about what this shows. The decision prompt prints the catastrophe probability verbatim, so the probe is a retrieval task, and these models copy printed values at ceiling throughout our data. It establishes that the risk was legible in the context the agent decided from - which is worth establishing, since it rules out the possibility that the number was lost or garbled - but it does not show that the agent represented the risk as a decision-relevant quantity, and it cannot distinguish an agent that weighed the risk and set it aside from one that never entered it into a calculation at all.

(f) Reading is not adding, and adding scales

The only comprehension axis on which the models struggled was *State*, the current cumulative situation, and the reason is specific and revealing. When we asked for a quantity that was printed verbatim in the prompt - for example, a player's own remaining money - accuracy was high for every model: under the same hidden-pool prompt used for the comparison below, five of the seven answered at or within a fraction of a point of ceiling, and Qwen2.5-7B scored 89.0%. The exception is Llama-3.1-8B, at 68.8% overall and only 46.0% in English: that model cannot reliably read the printed value either, so it is a poor test of the dissociation, and the contrast below should be read from the other six. When we asked for the same kind of quantity that had to be *summed* across rounds, namely the running pool total while it was hidden from the prompt, small models collapsed: Qwen2.5-7B answered correctly only 14.3% of the time and Llama-3.1-8B only 9.8%, even though both could read a printed value almost perfectly (figure 6(a)). That the failure is arithmetic rather than perceptual is confirmed by the A/B manipulation: when the same pool total was printed rather than hidden, accuracy jumped to 92–100% across the board. Reading, in other words, is not adding.

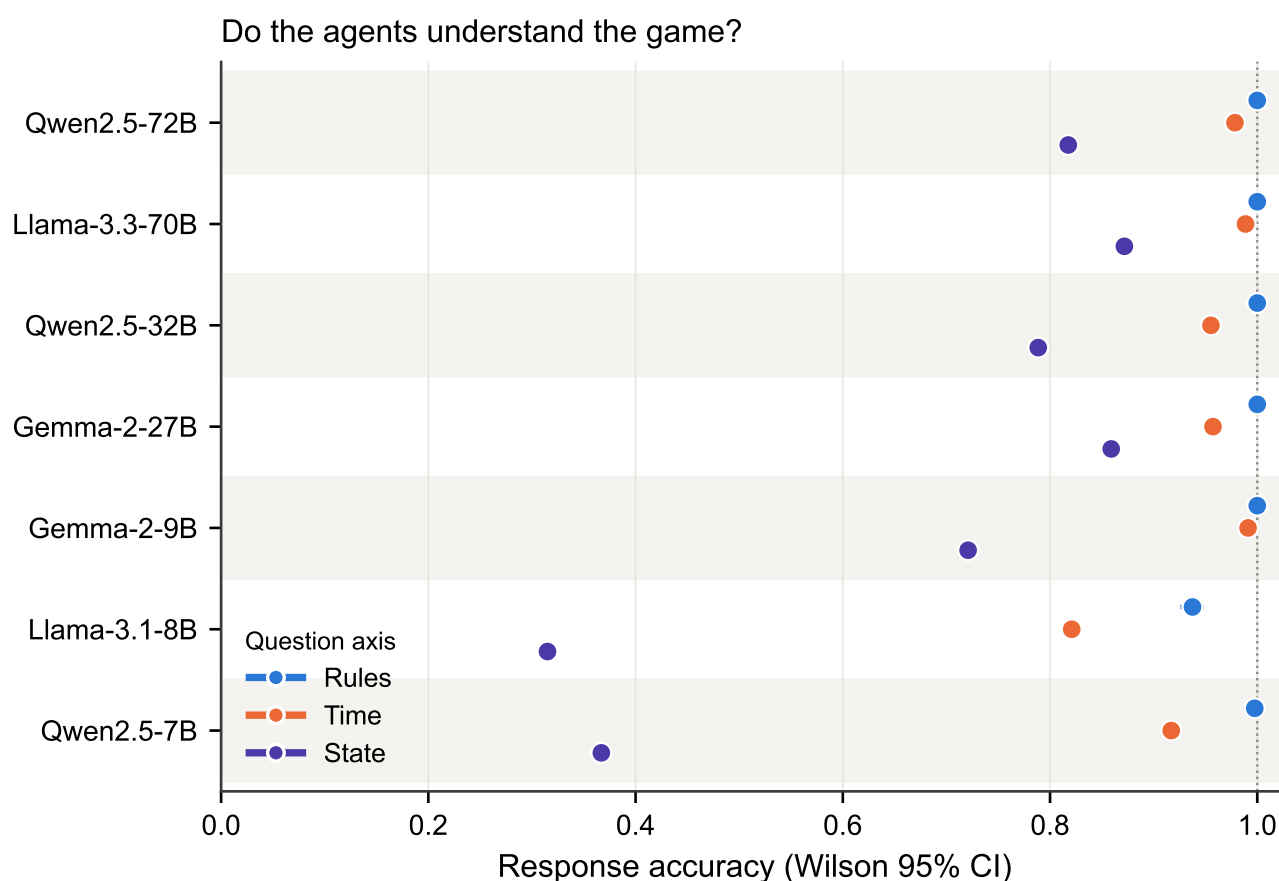


Figure 5. Comprehension by question axis. Response accuracy (dots) with Wilson 95% confidence intervals for each model, ordered by size (smallest at the bottom), on the Rules, Time and State axes; alternating bands group the three dots belonging to one model, and with 2,880–28,200 probes per point the intervals are narrower than the markers. Rules is at ceiling and Time is high (82–99%) for all models; State (cumulative situation) is the only axis that varies widely, and it improves broadly with scale.

This cumulative-addition deficit is a property of small models that scale largely removes. Accuracy on the hidden pool total rose steeply, if not perfectly monotonically, with size: 14.3% and 9.8% for the 7B and 8B models, 21.2%, 61.5% and 43.5% for the 9B, 27B and 32B models, and 81.7% and 88.2% for the 70B and 72B models (figure 6(b), lower curve); the two local reversals fall between families rather than within them. The juxtaposition in figure 6(b) is the crux of the paper: knowledge of the risk is flat and near-perfect across the whole size range, while the ability to track cumulative state climbs steeply with scale. By 70–72B the models both know the risk and can add the pool - and they still ignore the risk.

(g) Language preserves knowledge but degrades arithmetic

Running every condition in English and Vietnamese revealed a clean dissociation (figure 7). Behaviourally, the two languages produced identical target-reach rates for five of the seven models. Qwen2.5-32B dipped slightly (100% in English versus 93.3% in Vietnamese, and its single catastrophe fell in Vietnamese), and Llama-3.1-70B collapsed outright, from 100% in English to 0% in Vietnamese, so that its overall 50% was entirely a language split rather than a risk effect. In comprehension, language left rule and risk knowledge essentially untouched - Rules accuracy differed by under one percentage point between languages and the risk probability was read correctly in both - but it selectively attacked the arithmetic. State accuracy fell from 73.6% in English to 61.8% in Vietnamese, an almost twelve-point drop. The damage was concentrated in the middle of the range rather than at the bottom: Gemma-2-9B lost the most (87.8% to 56.4%, 31.5 points), whereas the two models that were already worst at cumulative addition had the least left to lose (Qwen2.5-7B 13.6 points, Llama-3.1-8B 11.3 points), and Llama-3.3-70B was unaffected (Vietnamese in fact 0.8 points higher). Language, then, changes what the models can compute far more than what they know.

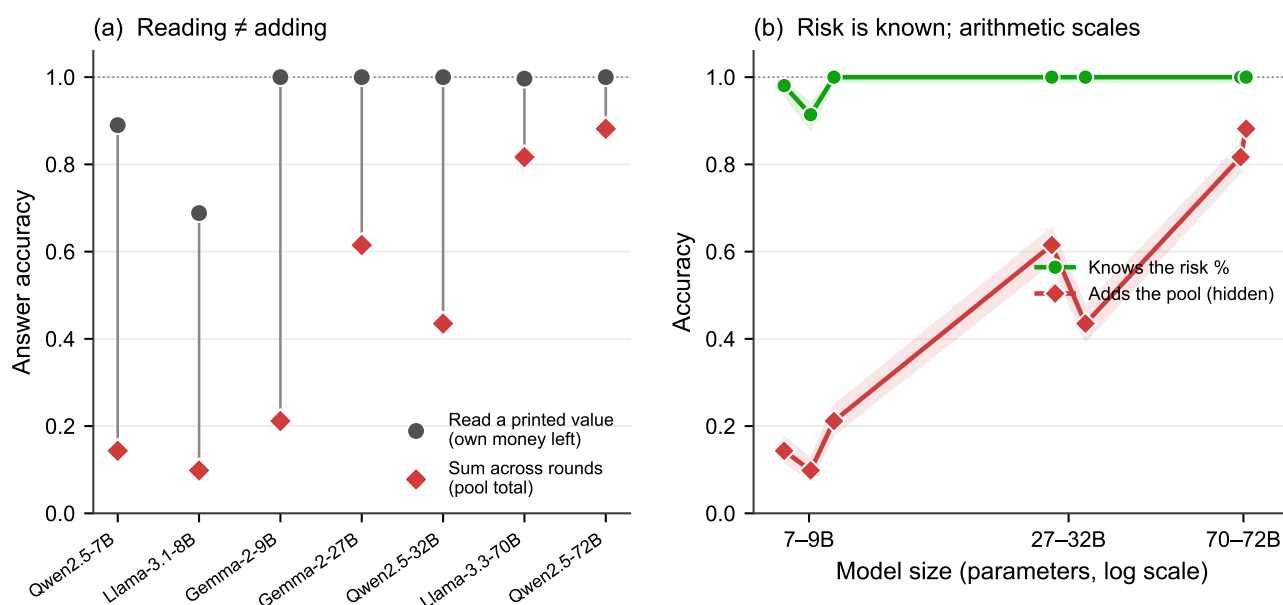


Figure 6. The gap is arithmetic, and it closes with scale. (a) Under the default hidden-pool prompt, accuracy on a value the agent can read straight off the prompt (its own money left, grey circles) versus a value it must sum across rounds (the pool total, red diamonds). Small models can read but cannot add. The same-question control is the A/B in the text: printing the pool total lifts accuracy on that identical question to 92–100%. (b) Knowledge of the catastrophe probability (green) is flat and near-perfect across the size range, whereas the ability to sum the hidden pool (red) rises steeply with scale. Even where both are near-ceiling, behaviour remains risk-insensitive. Bands are Wilson 95% intervals.

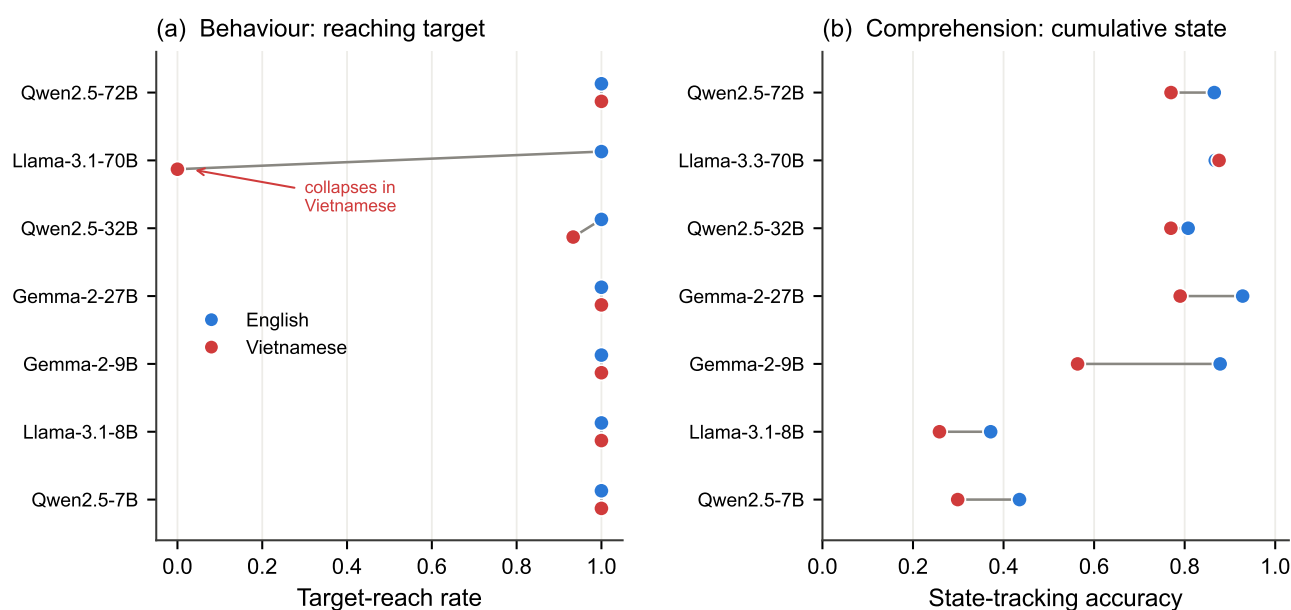


Figure 7. English versus Vietnamese. (a) Target-reach rate in each language, the two points of a pair offset vertically so that coincident values stay visible. Behaviour is identical across languages for five of the seven models; Qwen2.5-32B dips slightly (100% to 93.3%) and Llama-3.1-70B collapses from 100% to 0% in Vietnamese. (b) Cumulative-state accuracy in each language. Vietnamese selectively degrades the arithmetic; the largest loss is Gemma-2-9B (31.5 points), while the weakest adders have least left to lose and Llama-3.3-70B is unaffected.

4. Discussion

The central result of this study is a comparison of two manipulations measured under one prompt. Humans in this dilemma cooperate more when the catastrophe is more likely; across an order of magnitude of scale, no model did so to any consequential degree, the largest response being 8.8 points of 240. Yet the same models shifted by up to 95.3 points

Table 1. Headline results per model. Behavioural quantities pooled over risk and language (60 games/model): target-reach rate, group contribution (of 240) and realised catastrophe rate. Comprehension quantities from the in-situ probes: accuracy at stating the catastrophe probability (*risk knowledge*) and at summing the hidden pool (*cumulative arithmetic*). Comprehension used the Llama-3.3 revision at 70B. Human target-reach rates rise from 0% to 50% with risk.

Model	Size	Behaviour			Comprehension	
		Reach	Contribution	Catastrophe	Risk knowledge	Cumulative arithmetic
Qwen2.5-7B	7B	100%	232.7	0%	98.1%	14.3%
Llama-3.1-8B	8B	100%	185.4	0%	91.4%	9.8%
Gemma-2-9B	9B	100%	128.9	0%	100%	21.2%
Gemma-2-27B	27B	100%	126.3	0%	100%	61.5%
Qwen2.5-32B	32B	96.7%	127.6	1.7%	100%	43.5%
Llama-3.1-70B	70B	50%	121.0	36.7%	100%*	81.7%*
Qwen2.5-72B	72B	100%	132.5	0%	100%	88.2%

*Llama-3.3-70B in the comprehension study.

when the prompt merely displayed a number they already had the information to compute. Incentives moved them a little; salience moved them a lot. Whatever else this shows, it is not a failure of retrieval. The comprehension probes show that the models know the rules, can state the exact catastrophe probability of the game in front of them, and - at scale - can reconstruct its cumulative state. Table 1 makes the point in one line for Qwen2.5-72B: a model that answers the risk question perfectly and adds the hidden pool correctly 88.2% of the time nonetheless pools the same amount whether the risk of ruin is one in ten or nine in ten.

We are careful about what this gate does and does not establish. The battery tests whether a model can report the rules, read the history and sum the state; it does not test whether the model can put p to work decision-theoretically - no question asks for the expected payoff of missing the target at this p , or which contribution maximises it. Our results therefore rule out the explanation that the models misread the game, but they do not distinguish a model that has weighed the risk and set it aside from one that recalls the number without ever entering it into a calculation. Both are dispositions rather than misunderstandings, and the distinction between them is the natural target of a follow-up study. A second caveat is one of coverage: the comprehension panel used Llama-3.3-70B where the behavioural panel used Llama-3.1-70B, so the single model with genuine behavioural variance has no comprehension data of its own, and the joint claim rests principally on Qwen2.5-72B, with Gemma-2-27B and Qwen2.5-32B at the middle of the range.

If the agents are not tracking risk, what are they tracking? The salience result suggests an unglamorous answer: the contents of the prompt. Our own prompt names the equal-split solution, and the models that converge on “just enough” converge on very nearly that number, with Gemma-2-9B in English reproducing it exactly in all thirty games. Adding one more number to the prompt - the running total - moves them again, and by much more. A parsimonious reading of the whole panel is that these agents are executing the most salient available instruction rather than solving a dilemma, and that their apparent cooperativeness is in part an artefact of what we told them. A second contributing factor is plausible but, on our data, unproven: instruction tuning. Every model we studied was fine-tuned to be helpful and harmless [17, 18]; those procedures optimise for helpfulness and harmlessness rather than for cooperation as such, and we can only conjecture - not demonstrate here - that the resulting disposition generalises into a bias toward reaching a shared goal. Our data are at least consistent with such a tendency overriding the incentive structure of the game: the models converge on “reach the target” as a default and calibrate only how much private surplus to keep, not whether the target is worth reaching given the risk. This reading is reinforced by what scale does and does not change. Scaling from 7B to 72B improves a genuinely cognitive sub-skill, cumulative arithmetic, from near-chance to near-ceiling, exactly the kind of capability that larger models are known to acquire [19]; yet the same scaling leaves the behavioural prior untouched. Capability and the disposition to act on incentives are, here, decoupled - a caution for the assumption that more capable agents will automatically behave more like rational, risk-responsive decision-makers.

These findings connect to prior work whose picture is more mixed than a single summary allows. Some studies report LLMs that are markedly more cooperative and more forgiving than humans in repeated dilemmas [8, 9]; others report the opposite temperament, with GPT-4 retaliating unforgivingly after a single defection [10], and most models failing to sustain a shared resource without explicit institutional scaffolding [11]. Our result sits with the first group and sharpens it: in a dilemma structured by risk rather than by a payoff matrix, over-cooperation appears as an insensitivity to the very quantity that should govern it. Methodologically, the study extends the in-situ understanding-check paradigm from the two-player prisoner’s dilemma [8] to an N -player threshold game and shows that a single, well-chosen probe - can the model state the risk it faces? - separates one important confound, misreading, from a behavioural claim. We suggest that behavioural evaluations of LLM agents should routinely include such gates.

Our results carry three practical implications. First, LLM agents are unreliable surrogates for humans in risk-structured collective decisions: they fail to reproduce the human risk response, and would, for example, predict near-certain cooperation at a level of risk where human groups in fact reach the target only half the time. (The comparison is not uniformly unflattering: at $p = 0.9$ the least ceiling-bound model, Llama-3.1-70B, reaches the target in exactly half of its games, as human groups do; the divergence is concentrated at low risk, where the agents cooperate and humans do not.) Second, because the behavioural prior is invariant to scale while the underlying numeracy is not, one cannot infer an agent's incentive-responsiveness from its competence; more capable agents in a fleet may be quieter about, not freer of, a shared disposition. Third, language is not neutral: deploying the same model in a lower-resource language preserved its factual knowledge of the game but degraded its quantitative reasoning and, for a model living at the threshold, flipped its outcomes entirely. Multilingual deployment can therefore silently move quantitative behaviour even where surface comprehension looks intact.

Several limitations bound these conclusions, and the first is severe enough that we place it ahead of the others. Our decision prompt names the equal-split solution. An agent that contributes 2 is therefore consistent with a cooperative disposition, with plain instruction-following, or with anchoring on the only number the prompt offers, and this study cannot tell them apart. The evidence that this is not hypothetical is in our own data: Gemma-2-9B produced a group total of exactly 120.0 in all thirty English games, with zero variance, and Gemma-2-27B did the same in Vietnamese. Any claim in this paper about cooperative dispositions should be read as a claim about behaviour under a prompt that supplies a focal action; the clean experiment removes the hint and re-runs, and we regard that as the necessary sequel rather than an optional extension. For the same reason we have framed our central result as a comparison between two manipulations measured under the identical prompt - risk versus salience - since that contrast is internal and survives the confound. The groups were homogeneous and mute: all six agents were the same model and could not communicate, whereas human cooperation is shaped by heterogeneity and cheap talk, so our estimates speak to a specific, controlled regime rather than to open multi-agent interaction. The catastrophe was a single end-of-game lottery, which makes realised disaster rates noisy and is why we treated reach rate and contribution as the primary dependent variables. The baseline prompt did not print the running pool total, a defensible but not verbatim reading of the human display; the A/B manipulation was included precisely to measure the effect of that choice, and it isolates arithmetic rather than confounding it. Two confounds bound the scaling claim in particular. The 70B and 72B models were served with activation-aware weight quantisation while the smaller models were not, so "scale" is not a perfectly clean manipulation; and the Milinski design is old and widely reproduced, so it is likely present in every model's training corpus, meaning an agent may partly be recognising a familiar task rather than reasoning about it afresh. We studied instruction-tuned models only, so the prior we describe is a property of models as deployed and may differ for base models; without base-model controls the attribution to instruction tuning remains a conjecture. The comprehension panel used the Llama-3.3 revision at 70B where the behavioural panel used Llama-3.1, and we have kept the two labelled distinctly. Finally, we examined a single dilemma with three model families; whether the same disposition governs other risk-structured games, and whether it can be moved by targeted prompting or fine-tuning, are open questions.

Those questions define the natural next steps, and one of them we can already answer. We had expected that forcing explicit accounting - printing the running total each round - would repair the small models' arithmetic without touching their behaviour, on the reasoning that the risk they must respond to is not the state they must add. That expectation was wrong, and informatively so: printing the total changed contributions more than any incentive manipulation in the study. The remaining steps are to vary the prompt systematically rather than one factor at a time, in particular to remove the equal-split hint and re-run; to compare base and instruction-tuned checkpoints, without which the attribution to instruction tuning stays a conjecture; and to introduce heterogeneous groups and communication, which would test whether anything here survives social pressure.

5. Conclusion

Seven instruction-tuned language models, from 7 to 72 billion parameters and across three families, played a replication of the collective-risk social dilemma. None reproduced the human response to risk: no model reached the collective target more often when catastrophe was more likely, and although a correctly paired analysis shows a real positive risk response in three models, the largest is 8.8 of a possible 240 and changes no outcome. The decisive comparison is internal: printing the running pool total in the prompt, a manipulation carrying no incentive information whatever, moves contributions by up to 95.3 points in the same models under the same conditions. These agents track what the prompt makes salient far more closely than what the game makes costly. Comprehension probes bound the explanation - the agents recite the rules and read printed quantities at ceiling, while cumulative summation is a small-model gap that closes with scale - but they cannot establish that risk was ever treated as a decision variable, since the prompt prints it. Our own prompt also names the equal-split solution, and some models reproduce it exactly, so we cannot separate cooperative disposition from compliance with a supplied focal point. The methodological conclusion is the robust one: before a behavioural regularity in LLM agents is attributed to a preference, it should be tested against a salience control that changes the prompt without changing the incentives.

Ethics.

Data Accessibility.

Authors' Contributions.

Competing Interests.

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